Lucy Huang

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EDUCATION

***B.S. in Electrical Engineering*, University of California, Los Angeles** **Expected Graduation June 2029**

- Courses: Differential Equations, Data Structures and Algorithms, IoT, Multivariable Calculus, Discrete Math, Linear Algebra
- Clubs: IEEE WRAP (Wireless, RF, Analog Project), Engineering Student Association Intern, BruinSpace (RAPID Electrical), Design Build Fly (Electronics and Propulsion Team), Society of Women Engineering

SKILLS

- Languages and Software: C++, Python, Java, MATLAB, Linux, Ansys Lumerical, Autodesk Inventor, Fusion360
- Hardware/Lab Instrumentation: Oscilloscopes, Function Generator, UV-Vis-NIR spectroscopy, SEM, FTIR, Raman

RESEARCH EXPERIENCE

Quantum Light-Matter Cooperative (QLMC), UC Los Angeles, CA **September 2026 - Present**

Undergraduate Researcher [Faculty Advisor: Dr. Sergio Carbajo]

- Integrated dual comb spectroscopy on photonic chips for precision quantum sensing to maximize signal-to-noise ratio

Nanoscale Interfaces & Assembly Lab, UC San Diego MRSEC, La Jolla, CA **June 2024 - September 2026**

Student Researcher and Mentor [Faculty Advisor: Dr. Andrea R. Tao]

- Assembled gold nanospheres functionalized with graphene oxide, analyzed plasmonic coupling
- Synthesized copper(I) oxide nanocrystals, CAD intersectional structures for finite-difference time-domain (FDTD)
- Developed FDTD automation script with Lumerical Python API, to optimize data collection process.

WORK EXPERIENCE

***Math and Language Arts Teaching Assistant,* Art of Problem Solving, San Diego, CA** **April 2026 - Present**

- Directed Virtual Homework Help sessions on topics ranging from Pre-Algebra to High School Contest Math
- Led in-person summer camps for Math Beasts Camp 3 and High School Contest Math

***Court Monitor,* University City Racquet Club, San Diego, CA** **June 2023 - February 2025**

- Responsible for business operations: court bookings and retail sales, tournaments and directed weekly league matches, membership enrollment, coordinated facility repairs and maintenance, consistent and exceptional customer service

PUBLICATIONS, HONORS & AWARDS

Research

- **Huang, L.**, et al. (2026, August) *Co-Assembly of Au Nanospheres and Graphene Oxide into Plasmonically Active Films.* 2026 Summer Research Conference at UC San Diego
- **Huang, L.**, et al. (2025, November) *Simulating the Optical Properties of Copper(I) Oxide Nanocubes in Monolayer Films and with Structural Alterations*.
  - 2026 California Science and Engineering Fair: Honorable Mention in Physics and Astronomy
  - 2026 Greater San Diego Science and Engineering Fair: 1st Place Award in Senior Engineering: Energy, Materials, and Transport, Winner of General Atomics Sciences Education Foundation, DRS Daylight Solutions
  - 2025 Regeneron Science Talent Search: Research Manuscript
  - 2025 MRSEC RIMSE Poster Session at UC San Diego: Most Engaged High School Researcher
- **Huang, L.**, et al, (2024, April) *Morphological Changes and Kinetic Insights of Copper(I) Oxide Nanocrystals*.
  - 2025 California Science and Engineering Fair: Finalist in Chemistry
  - 2025 Greater San Diego Science and Engineering Fair: 1st Place Award in Senior Chemistry, Winner of Office of Naval Research Award, Association for Women in Science, American Society of Materials International
  - 2024 MRSEC RIMSE Poster Session at UC San Diego: Most Engaged High School Researcher

Science Olympiad

- *Southern California States 2024-2026* – 5+ individual awards
- *San Diego Regional Finals* 2019-2026 – led school's top 5 ranking and 10+ awards
- *Invitationals: Golden Gate 2026, Triton (UCSD) 2025-2026*, *Stanford 2025*, *San Diego County 2019* – 10+ awards

NCWIT Aspirations in Computing Rising Star Award, January 2026

LEADERSHIP

***Co-President, Head Coach, Event Coach, Competitor,* Science Olympiad** **September 2018 - May 2026**

- Led 30+ members per year, improving state rankings by 3 spots. Coached to invitational, regional, and state victories

***Student Leadership Board Member,* Greater San Diego Science and Engineering Fair** **May 2025 - May 2026**

- Promoted GSDSEF to nearby schools, led workshops and mentored students from project design to execution

***Mentee Representative,* UCSD Empowering Development for Girls in Engineering (EDGE)** **April 2023 - May 2026**

- Elected and served on the board, advocating mentee interests. Committed 4 years to bi-weekly engineering program